

HYBRID HIDDEN MARKOV MODELING FOR TARGET RECOGNITION BASED ON HIGH-RESOLUTION RADAR SIGNAL

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Abstract

Hidden Markov Model (HMM) is a commonly used modeling approach for the radar target recognition based on high-resolution range profile (HRRP). The HMM describes HRRP with the assumption of Gaussian observation probability, which results in the inaccurate description due to the non-Gaussian property of HRRP. To this end, we propose a hybrid HMM framework, which uses an extra model to fit the observation probability rather than making distribution assumptions. The extra model estimates the posterior probability of the state, and then can calculate the pseudo observation probability, which in turn modulates the other model parameters in HMM. The interplay between HMM and the extra model ultimately leads to a better description for observed data. In this paper, we use the support vector machine (SVM) and deep neural network (DNN) separately as the extra model and validate the superiority of our hybrid HMM in HRRP-based target recognition.

1 Introduction

With the intelligent development of the modern war, there is an increasing demand for radar signal-based target recognition. With the consideration of easy acquisition and processing, many researchers choose the high-resolution range profile (HRRP) of the target to complete the recognition task [1], [2]. As shown in Fig. 1, the target can be considered as composed of scatterers located in many range cells along the radar line-of-sight (LOS), and the HRRP is a real-valued vector with each element being the amplitude of the summation of complex echoes from scatterers in the corresponding range cell. Since the scatterer distribution is the reflection of target structure (such as target size and shape) and different targets have different scatterer distributions, the HRRP can thus be used for the effective target recognition. References [1]-[6] realize the HRRP-based radar target recognition by exploiting the parametric statistical models, of which [4]-[6] successfully utilize the hidden Markov model (HMM) for modeling HRRPs. HMM, a generative probabilistic model, considers the observation sequence being generated by unobserved (hidden) states, where the Markov chain for the transitions between two hidden states holds. In [4], the HMM is utilized to characterize HRRP's waveform constituents extracted by the relax algorithm; the moments

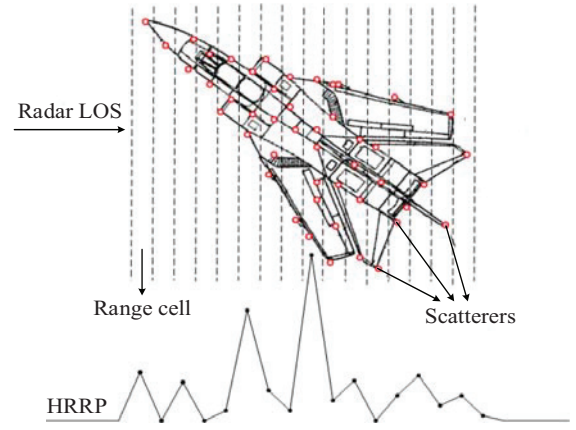


Fig. 1. Illustration of an HRRP sample from a plane target, where the circles on the plane represent the scatterers.

from HRRP are modeled via the nonstationary HMM in [5]. Rather than modeling the features of HRRP which often leads to the information loss, reference [6] views an HRRP sample as a sequence and each sample is modeled via an HMM, wherein a range cell is considered as an observation in the sequence. This approach retains useful target information for recognition while assuring the low-dimensional modeling of HMM, which avoids the higher model complexity. One of the characteristics of above method is that the observation probability, i.e., $p(x|s)$ (x and s denote the observation and hidden state, respectively), is assumed as a Gaussian distribution, referred to as the HMM-Gaussian model. However, Du et al. [2] find out that the statistical distribution of the echo from each range cell is in fact not strictly Gaussian distributed. Therefore, there is a great challenge for the HMM-Gaussian model to accurately describe HRRP's statistical characteristics, which could weaken the recognition performance eventually. Considering the non-Gaussian property of HRRP echoes, this paper uses an additional model to fit the observations without making assumptions any more. In detail, the additional model produces the posterior probability over state $p(s|x)$ with taking the state indicator inferred by HMM as the ground truth. Due to the Bayes theorem of $p(x|s) \propto p(s|x)/p(s)$, the observation probability can thus be converted into the

ratio of state posterior to state prior which can be estimated via the HMM. Afterwards, the calculated observation probability is in turn used to adjust the estimations of model parameters in HMM. The HMM and additional model are optimized iteratively until convergence, which ultimately leads to the better data fitting performance than the traditional HMM. We call above modeling method as the hybrid HMM and specially take the support vector machine (SVM) and deep neural network (DNN) as the additional model in this paper.

The rest of this paper is organized as follows. The HMM-Gaussian model for HRRP is reviewed in Section 2. The overall structure of the hybrid HMM is introduced in Section 3.1. Then, SVM and DNN-based hybrid HMMs are presented in Section 3.2. Section 4 conducts the experiments on measured HRRP data, and Section 5 concludes this paper.

2. HMM-Gaussian modeling for HRRP

The HMM represents the generative process from underlying states to observed sequence data [7]. For an sequence observation with its length being L , i.e., $\mathbf{x} = (x_1, x_2, \dots, x_L)$, the HMM-Gaussian model represents the generation of $\mathbf{x}$ as $p(x_i | s_i) = \mathcal{N}(x_i | \theta_{s_i})$, where $\mathcal{N}(\cdot)$ denotes the Gaussian distribution, s_i denotes the hidden state of x_i with $s_i \in \{1, \dots, K\}$ and K being the K th state, θ_{s_i} denotes the mean and variance of the Gaussian distribution corresponding to s_i and a Normal-Gamma prior H is often imposed on the state. For the value of the state, it depends on state transition probability $w_{i,j}$ with $w_{i,j} = p(s_{i+1} = j | s_i = i)$ ($i, j \in \{1, \dots, K\}$) and the initial state distribution $w_{0,i} = P(s_1 = i)$.

During the HMM-Gaussian modelling, the target-aspect sensitivity of the HRRP [8] should be firstly processed. Usually, we adopt the aspect-frame partition method. In detail, the whole HRRP dataset is divided into several subsets, in each of which the motion through range cells (MTRC) of scatterers corresponding to any HRRP does not occur. Afterwards, we call each HRRP subset a frame. In [6], an HMM-Gaussian model is constructed for each HRRP frame and all frames share a large set of states, i.e., the distribution parameters corresponding to each state are shared among all frames, from which the needed states for each frame is automatically selected via the truncated stick-breaking prior [9]. For the n th HRRP observation sequence from the m th frame of target c , i.e., $\mathbf{x}^{(c,m,n)} = [x_1^{(c,m,n)}, \dots, x_L^{(c,m,n)}]^T$ ($n \in \{1, \dots, N\}$, $m \in \{1, \dots, M_c\}$, $c \in \{1, \dots, C\}$ with N , M_c , and C denote the number of observed sequences in each frame, the number of frames in the c th target, and the number of targets, respectively), the HMM-Gaussian model is mathematically represented as

$$x_l^{(c,m,n)} \sim \mathcal{N}\left(\theta_{s_l^{(c,m,n)}}\right), \theta_k \sim H, s_l^{(c,m,n)} \sim \begin{cases} \mathbf{w}_0^{(c,m)}, & \text{if } l = 1 \\ \mathbf{w}_{s_{l-1}^{(c,m,n)}}^{(c,m)}, & \text{if } l \geq 2 \end{cases}$$

$$\mathbf{w}_i^{(c,m)} \sim \text{GDD}\left(\mathbf{1}_{(K-1) \times 1}, [\beta_i]_{(K-1) \times 1}\right), \beta_i \sim \text{Gamma}(a_\alpha, b_\alpha), \quad (1)$$

where $s_l^{(c,m,n)}$ denotes the state indicator of the l th observation in the n th HRRP sequence from the m th frame of target c . The i th ($i = 1, \dots, K$) row of the state transition matrix $\mathbf{W}^{(c,m)}$, i.e., $\mathbf{w}_i^{(c,m)} = [w_{i,1}^{(c,m)}, \dots, w_{i,K}^{(c,m)}]$, is imposed on a truncated stick-breaking prior with $\text{GDD}(\cdot)$ denoting the general Dirichlet distribution [10], $\mathbf{1}_{(K-1) \times 1}$ and $[\beta_k]_{(K-1) \times 1}$ denoting the $(K-1)$ -length vector of one and β_k , respectively, and $\text{Gamma}(\cdot)$ representing the Gamma distribution. Via the variational Bayesian (VB) [11] algorithm, the model parameters, including the $s_l^{(c,m,n)}$, $\mathbf{W}^{(c,m)}$, and $\theta_{s_l^{(c,m,n)}}$, can be derived.

3 Hybrid HMM construction

3.1 The structure of the hybrid model

The HMM-Gaussian model assumes that observations in an HRRP sequence obey Gaussian distribution, which is not appropriate according to the analysis in [2]. Rather than making a specific distribution assumption, we estimate the observation probability by introducing an extra model, which combines with HMM-Gaussian and forms a hybrid model, i.e., hybrid HMM, in this paper. In our hybrid HMM, the extra model takes the observation $x_l^{(c,m,n)}$ as the input and outputs the posterior of state $p(s_l^{(c,m,n)} | x_l^{(c,m,n)})$ with the state indicator inferred by HMM-Gaussian being selected as the ground truth. According to Bayes theorem, we have

$$p(x_l^{(c,m,n)} | s_l^{(c,m,n)}) \propto p(s_l^{(c,m,n)} | x_l^{(c,m,n)}) / p(s_l^{(c,m,n)}). \quad (2)$$

Eq. (2) makes it possible to fit the non-Gaussian HRRP, thus offering the potential to make a more accurate depiction for observed sequences.

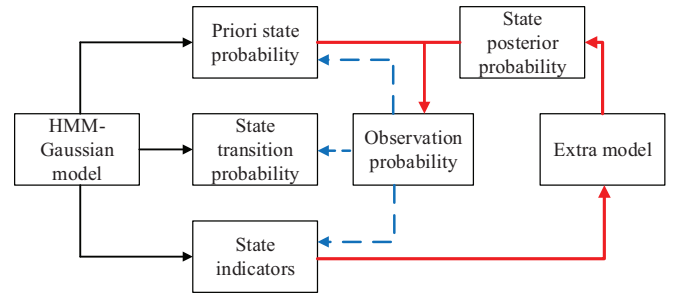


Fig. 2. Diagram of the hybrid HMM.

Fig. 2 presents the diagram of the proposed hybrid HMM. In the training stage, we first learn the HMM-Gaussian model, from which the state prior, the state transition matrix, and the state corresponding to each observation can be inferred, as indicated by black lines in Fig. 2. Then we train the extra model with the learned state indicator being taken as the ground truth, and obtain the state posterior probability with which the observation probability can be derived according to

(2), marked out in bold red lines in Fig. 2. Afterwards, the calculated observation probability is in turn used to modify the parameters of HMM-Gaussian (shown with blue dotted lines). The HMM-Gaussian and the extra model are optimized iteratively until convergence.

In the test stage, the statistical recognition method [6], [8] is adopted to predict the class label of a test HRRP sequence $\mathbf{x}^*$. By statistical recognition we mean that the class $\mathbf{x}^*$ belonging to has the maximum frame likelihood $p(\mathbf{x}^*|c, m)$, which can be expressed as

$$p(\mathbf{x}^*|c, m) = \sum_{s^*} \prod_{l=1}^L \frac{p(s_l^*|x_l^*)}{p(s_l^*)} \prod_{l=2}^L p(s_l^*|s_{l-1}^*) p(s_1^*) \prod_{l=1}^L p(x_l^*), \quad (3)$$

with the parameters corresponding to $\mathbf{x}^*$ being inferred via the learned hybrid HMM. In the following, we take the SVM and DNN as examples to demonstrate the acquisition of the state posterior.

3.2 Hybrid HMM-SVM and hybrid HMM-DNN

SVM is a binary classifier that constructs a hyperplane to divide a problem region into two classes. It can be mathematically expressed as

$$\begin{aligned} \min \quad & -\sum_{i=1}^N \alpha_i + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \\ \text{s.t.} \quad & \sum_{i=1}^N y_i \alpha_i = 0, \quad 0 \leq \alpha_i \leq C, \end{aligned} \quad (4)$$

with $K(\cdot, \cdot)$, y_i , α_i , b , C being the kernel function, label of $\mathbf{x}_i$, Lagrange multiplier, bias, tunable constant, respectively, and outputs the distance measure from data point to decision hyperplane, i.e., $t(\mathbf{x}) = \sum_{i=1}^N \alpha_i y_i K(\mathbf{x}, \mathbf{x}_i) + b$.

An important issue required to be addressed in our hybrid HMM-SVM model is the conversion from the distance measure to posterior probability. Platt [12] suggests a method that describes the probabilistic output as

$$p(y=1|\mathbf{x}) = \frac{1}{1 + \exp(A t(\mathbf{x}) + B)} \quad (5)$$

where the estimation of parameters A and B are obtained by optimizing the following cross-entropy loss

$$\min_{z=(A,B)} F(z) = -\sum_{i=1}^N (y_i \log(p_i) + (1-y_i) \log(1-p_i)). \quad (6)$$

Referring to (5), our hybrid HMM-SVM can obtain the state posterior $p(s_l|x_l)$ by considering x_l as the input of the SVM and the state indicator s_l learned from HMM-Gaussian as the label (ground truth) of s_l .

For the hybrid HMM-DNN, it exploits the DNN to estimates the posterior probability of the state. Considering a DNN with D layers, the output of each hidden layer can be expressed as $\mathbf{v}^{d+1} = f(\mathbf{W}^d \mathbf{v}^d + \mathbf{b}^d)$ ($1 \leq d \leq D-2$), where

$\mathbf{v}^d \in \mathbb{R}^{N_d \times 1}$ with N_d denoting the number of neurons in layer d and $\mathbf{v}^1 = \mathbf{x}$; $\mathbf{W}^d \in \mathbb{R}^{N_{d+1} \times N_d}$ and $\mathbf{b}^d \in \mathbb{R}^{N_{d+1} \times 1}$ are the weight matrix and the bias, respectively; $f(\cdot)$ is an

activation function, such as the sigmoid function used in this paper. Then the D th layer, i.e., output layer, of the DNN is $\mathbf{v}^D = g(\mathbf{W}^{D-1} \mathbf{v}^{D-1} + \mathbf{b}^{D-1}) = g(\mathbf{z})$ and $g(\cdot)$ is often selected as the softmax function in classification task, i.e., $v_i^D = e^{z_i} / \sum_{i=1}^{N_D} e^{z_i}$ with z_i the i th element of $\mathbf{z}$, which can approximately represent the posterior probability $p(\mathbf{v}^D|\mathbf{x})$.

The network parameters $\Theta = \{\mathbf{W}^d, \mathbf{b}^d\}_{d=1}^{D-1}$ can be obtained by minimizing the following loss

$$L = \frac{1}{M} \sum_{m=1}^M J(\Theta; \mathbf{x}_m, y_m) = \frac{1}{M} \sum_{m=1}^M y_m \log \mathbf{v}^D, \quad (7)$$

where y_m is the label of the m th input $\mathbf{x}_m$.

Similar to the hybrid HMM-SVM, we take the x_l as the input of the DNN and the state indicator s_l as the label of x_l , and thus the state posterior $p(s_l|x_l)$ can be calculated.

4 Experimental results

4.1 Data description and experiment setup

In this section, the performance of the proposed method is evaluated by applying our hybrid HMM to the measured HRRPs, which are from three real airplanes, including Yark-42, Cessna Citation S/II, and An-26 [2], [6], [8]. The radar center frequency and bandwidth in the experiment are 5520MHz and 400MHz, respectively. Moreover, there are 675 training HRRPs and 5200 test HRRPs with each HRRP sequence has 128 observations. In addition, considering the target-aspect sensitivity of HRRP, 35 HRRP frames are obtained for the training dataset of Yark-42 via the aspect-frame partition method, similarly 50 frames for that of Cessna Citation S/II and 50 frames for that of An-26, i.e., $C=3$, $M_1=35$, $M_2=M_3=50$, $N=5$. In addition, centroid alignment [13] and L_2 norm are used to handle HRRP's time-shift and amplitude-scale sensitivities.

In the experiment, the total number of hidden states is set to 20. For the hybrid HMM-SVM, we select the radial basis function (RBF) as the kernel function. For our hybrid HMM-DNN, the architecture of DNN is 1-50-40-20 and is pre-trained with the restricted boltzmann machine (RBM) [14].

4.2. Recognition performance and results analysis

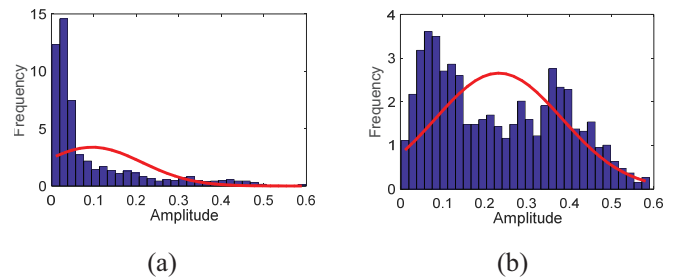


Fig. 3. Statistical distributions of echoes from two typical range cells of HRRP samples. (a) The the long-tailed distribution, (b) The bimodal distribution.

To analyze the HRRP distribution, we present the statistical histograms of echoes from some typical range cells in Fig. 3, where the red curve is the fitting result via Gaussian probability density function.

Obviously, the above two statistical characteristics cannot be accurately fitted by the Gaussian distribution (the long-tailed distribution in Fig. 3(a) and bimodal distribution in Fig. 3(b)). Therefore, HMM-Gaussian model is not strictly suitable for the description of HRRP, and the proposal of our hybrid HMM is of great significance.

Fig. 4 gives a comparison on the recognition performance of the hybrid HMMs with other related models, including SVM classifier, the principal component analysis (PCA) model [8], probability principal components analysis (PPCA) model [8], factor analysis model with multi-task learning (MTL-FA) [15], and HMM-Gaussian model. If we take SVM as the benchmark, the PCA, PPCA and MTL-FA models has the lower recognition accuracies because of their Gaussian representation for HRRP and the higher degree of model complexity. For the HMM-Gaussian, its modeling approach for each range cell significantly reduces the number of estimating parameters, thus having a better recognition performance than above three models. However, due to the inaccurate description of HRRP for HMM-Gaussian, its performance is lower than those of the HMM-SVM and HMM-DNN.

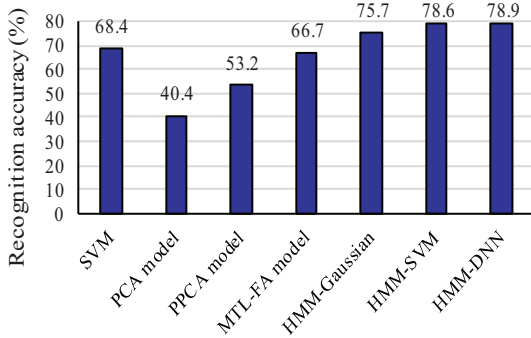


Fig. 4. Recognition accuracies of the hybrid HMM models and other related models.

In Fig. 5, we present the usage of states for HMM-based models. There are only about half of the states being selected, and thus the setting of 20 initial states is large enough to fully represent observations. Moreover, there are some differences in state distribution between the hybrid HMM-based models and HMM-Gaussian model, which demonstrates that the extra model (SVM and DNN) really works in modifying observation probability.

In Fig. 6, we analyze the effect of initial state number on the model recognition ability. When the number of the initialized states is too small to describe the complexity of HRRP data, the model's recognition performance will be poor, and however, it will increase the number of parameters to be learned when the number of initialized states is set too large, thus leading to the higher model complexity and ultimately the lower recognition accuracy. As shown in Fig. 6, the setting of 20 initial states is a better choice for HRRPs used in our experiment.

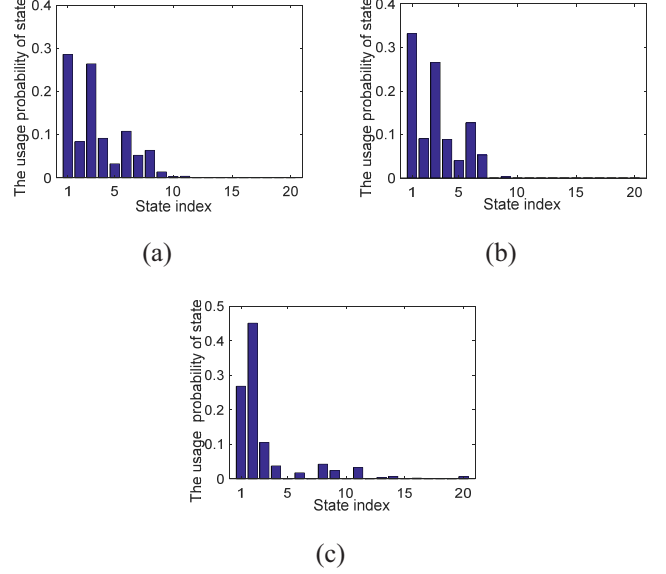


Fig. 5. The usage of hidden states for different HMM-based models. (a) HMM-Gaussian, (b) HMM-SVM, (c) HMM-DNN.

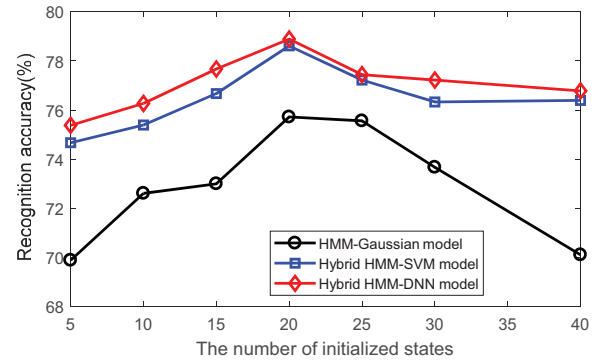


Fig. 6. Variations of recognition accuracy with the number of initialized states for different HMM-based models.

5 Conclusion

The hybrid HMM framework is proposed for modeling HRRP of radar target. In the hybrid HMM, an extra model is introduced to estimate the observation probability of HRRP sequence, instead of making Gaussian distributed assumption in the traditional HMM. Due to the typical non-Gaussian property of HRRP, the hybrid HMM can fit the observation more precisely, and thus, its modeling accuracy is significantly improved. In addition, we specially use the SVM and DNN as the extra models and illustrate the advantages of the proposed hybrid HMM-based models in HRRP-based target recognition than other related models.

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